Sum of Subarray Minimums (medium)

Problem Statement

Given an integer array `arr`, find the `sum` of the `minimum` elements of all possible `subarrays` of `arr`. Return the answer modulo $10^9 + 7$ as it may be very large.

Examples

Example 1:

- **Input**: [2, 3, 1]
- **Expected Output**: 10
- **Justification**: The subarrays are [2], [3], [1], [2, 3], [3, 1], and [2, 3, 1]. Their minimums are **2**, **3**, **1**, **2**, **1**, and **1** respectively. Summing these minimums yields **10**.

Example 2:

- **Input**: [4, 5, 2]
- **Expected Output**: 19
- **Justification**: The subarrays are [4], [5], [2], [4, 5], [5, 2], and [4, 5, 2]. Their minimums are **4**, **5**, **2**, **4**, **2**, and **2** respectively. Summing these minimums gives us **19**.

Example 3:

- **Input**: [7, 2, 8, 4]
- **Expected Output**: 35
- **Justification**: The subarrays are [7], [2], [8], [4], [7, 2], [2, 8], [8, 4], [7, 2, 8], [2, 8, 4], and [7, 2, 8, 4]. Their minimums are **7**, **2**, **8**, **4**, **2**, **2**, **4**, **2**, **2**, and **2** respectively. Summing these minimums totals **35**.

Try it yourself

Try solving this question here:



```
1   import java.util.Stack;
2
3   public class Solution {
4       public int sumSubarrayMins(int[] arr) {
5           // ToDO: Write Your Code Here.
6           return 0; // Return the final sum.
7       }
8   }
9
```

Test Case | Run Code Result | Stdout

```
[2, 3, 1]
```

Use Example Testcases | Run | Submit

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Solution: Delete Node In A BST

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Solution: Sum Of Subarray Minimums

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